Lebesgue Integral - II

This section contains the following topics:

- Uniform Integrability
  - Vitali Convergence Theorem
  - General Vitali Convergence Theorem
- Convergence in Measure
- Characterization of Riemann and Lebesgue Integrability

Uniform Integrability: Vitali Convergence Theorem

The Vitali Convergence generalizes Lebesgue Dominated Convergence. In order for $\lim \int_E f_n = \int_E \lim f_n$, Lebesgue Dominated Convergence Theorem requires that $|f_n| \leq g$ where g is integrable, whileVitali Convergence Theorem states that if $\{f_n\}$ are uniformly integrable. As we will show, $|f_n| \leq g$ implies that f_n are uniformly integrable.

Lemma 1.11.1: Let $m(E) < \infty$. Then E is the disjoint union of a finite collection of measurable subsets $\bigcup_{k=1}^{N} E_k$, where $m(E_k) \leq \delta$ for any $\delta > 0$.

Proof: Since $m(E) < \infty$, we must have

$$\lim_{n\to\infty} m(E - [-n, n]) = 0$$

. Otherwise, there will be a $\delta > 0$ such that $m(E - [-n, n]) \geq \delta$ for infinitely many n, resulting in $m(E) = \infty$ which is a contradiction.

So for any $\delta > 0$ there exists some M such that $m(E - [-M, M]) < \delta$. Take $N = \lceil \frac{2M}{\delta} \rceil + 1$, and $\delta' = \frac{2M}{N}$. Clearly $\delta' < \delta$. Take $E_k = [-M + \delta' \cdot (k-1), -M + \delta' \cdot k)$ and $E_0 = E - [-M, M]$. Then $E = \bigcup_{k=0}^{N} E_k$ and each $m(E_k) < \delta$.

QED

The following lemma is one characterization of integrable function.

Lemma 1.11.2: If f is integrable over E, then for any $\epsilon > 0$ there exists a $\delta > 0$ such that for any measurable $A \subseteq E$ and $m(A) < \delta$, we have

$$\int_A |f| < \epsilon$$

Conversely, suppose $m(E) < \infty$. If for any $\epsilon > 0$ there is a $\delta > 0$ such that any measurable $A \subseteq E$ with $m(A) < \delta$ satisfies $\int_A |f| < \epsilon$, then f is integrable over E.

Proof: We may assume f is non-negative. The general case can be proved using the fact that $|f^+ - f^-| \leq |f^+| + |f^-|$.

Suppose f is integrable over E. Note f is non-negative. So there exists some bounded non-negative function h of finite support such that $0 \leq h \leq f$ and $\int_E f - \int_E h < \frac{\epsilon}{2}$. For any measurable subset $A \subseteq E$, we have

$$\int_A f = \int_A [(f-h)+h] = \int_A (f-h) + \int_A h \leq \int_E (f-h) + \int_A h = \frac{\epsilon}{2} + \int_A h$$

. Note that h is bounded. So there exists some M such that $0 \leq h \leq M$. So

$$\int_A f \leq \frac{\epsilon}{2} + \int_A h = \frac{\epsilon}{2} + M \cdot m(A)$$

Now choose a δ such that $\delta < \frac{\epsilon}{2M}$. Then $\int_A f < \epsilon$ for any A with $m(A) < \delta$.

Conversely, suppose $m(E) < \infty$ and the above condition holds. We would like to show that f is integrable. Choose $\epsilon = 1$ and let δ meet the challenge. By Lemma 1.11.1, $E = \bigcup_{k=1}^N E_k$, where $m(E_k) < \delta$. Then $\int_{E_k} f < 1$. So for any bounded non-negative function h of finite support where $0 \leq h \leq f$ on E, we have $\int_{E_k} h \leq \int_{E_k} f < 1$. Therefore, by finite additivity $\int_E h = \sum_{k=1}^N \int_{E_k} h < N$. This is true for any such function h. As a result, $\int_E f < N$ and thus is integrable over E.

QED

We say that a family of functions $\mathcal{F} = \{f\}$ is **uniformly integrable**, if every member is integrable and they have uniformly the same δ that meets the same challenge ϵ.

Formally, $\mathcal{F} = \{f\}$ on E is **uniformly integrable**, if for any $\epsilon > 0$ there exists a $\delta > 0$ such that for any measurable subset $A \subseteq E$ with $m(A) < \delta$, we have $\int_E |f| < \epsilon$ for every $f \in \mathcal{F}$.

Theorem 1.11.3 (Vitali Convergence Theorem): Assume $m(E) < \infty$. If $\{f_n\}$ is uniformly integrable over E, and $f_n \to f$ a.e., then f is integrable over E and

$$\lim_{n\to\infty} \int_E f_n = \int_E f$$

Proof: Take $\epsilon = 1$ and let δ meet the challenge. Since $m(E) < \infty$, $E = \bigcup_{k=1}^N E_k$ where $m(E_k) < \delta$. So $\int_E |f_n| = \sum_{k=1}^N \int_{E_k} |f_n| < N$ for every n. By Fatou's Lemma, we have

$$\int_E |f| \leq \liminf \int_E |f_n| < N$$

, and thus f is integrable over E.

To justify the exchange of $\lim$ and $\int$, note that for every n

$$
\begin{aligned}
\Big|\int_E f_n - \int_E f\Big| &\leq \int_E |f_n - f| \\
&= \int_{E-A} |f_n - f| + \int_A |f_n - f| \\
&\leq \int_{E-A} |f_n - f| + \int_A |f_n| + \int_A |f|
\end{aligned}
$$

Since f_n is uniformly integrable, for any $\epsilon > 0$, let δ meets the challenge of $\frac{\epsilon}{3}$, and choose A such that: (1) $m(A) < \delta$, and (2) by Egoroff's Theorem, $f_n \to f$ uniformly on $E - A$.

Since $m(A) < \delta$, we have $\int_A |f_n| < \frac{\epsilon}{3}$ for every n. By Fatou's Lemma, $\int_A |f| < \frac{\epsilon}{3}$.

Since $f_n \to f$ uniformly on $E - A$, there exists some N such that when $n \geq N$ we have $|f_n - f| < \frac{\epsilon}{3 \cdot m(E)}$. So $\int_{E-A} |f_n - f| < \frac{\epsilon}{3 \cdot m(E)} \cdot m(E) = \frac{\epsilon}{3}$.

Therefore, when $n \geq N$, $|\int_E f_n - \int_E f| < \epsilon$.

QED

Vitali Convergence require uniform integrability of $\{f_n\}$. The following propositions provide two such sequences.

Proposition 1.11.4: $\{f_n\}$ is a sequence of measurable functions on E. If there is a function g such that $|f_n| \leq g$ and g is integrable over E, then $\{f_n\}$ is uniformly integrable.

Proof: Since g is integrable, by Lemma 1.11.2 there is some δ, meeting the challenge of ϵ, and for any $A \subseteq E$ with $m(A) < \delta$ we must have $\int_A |g| = \int_A g < \epsilon$. Since $|f_n| \leq g$, we have $\int_A |f_n| \leq \int_A g < \epsilon$ for every n. So $\{f_n\}$ is uniformly integrable.

QED

Proposition 1.11.5: $\{f_k\}_{k=1}^N$ is a finite sequence of measurable functions on E. Suppose every f_k is integrable over E, then $\{f_k\}_{k=1}^N$ is uniformly integrable .

Proof: For any $\epsilon > 0$, since every f_k is integrable over E, by Lemma 1.11.2 there is some δ_k meeting the challenge of ϵ for f_k. Now take $\delta = \min\{\delta_1, \ldots, \delta_N\}$, and δ is the uniform value that meets the challenge of ϵ for all f_k.

QED

Uniform Integrability: General Vitali Convergence Theorem

Vitali Convergence Theorem requires that $m(E) < \infty$. For a general E, we need one more condition to guarantee the convergence.

Lemma 1.11.6: If f is integrable over E, then for any $\epsilon > 0$ there is some subset E_0 with finite measure such that

$$\int_{E-E_0} |f| < \epsilon$$

Proof: Since f is integrable, so is $|f|$. Therefore, there is some non-negative bounded function of finite support $g \leq |f|$ such that $\int_E (|f| - g) \leq \int_E |f| - \int_E g < \epsilon$. Let E_0 be the finite support of g. Then $g(x) = 0$ for $x \in E - E_0$. So,

$$\int_{E-E_0} |f| = \int_{E-E_0} (|f| - g) \leq \int_E (|f| - g) < \epsilon$$

QED

We say that a family of functions $\mathcal{F} = \{f\}$ is **tight** provided that for any $\epsilon > 0$ there is a uniform subset E_0 with finite measure such that $\int_{E-E_0} |f| < \epsilon$ for every $f \in \mathcal{F}$.

Theorem 1.11.7 (General Vitali Convergence Theorem): $\{f_n\}$ is uniformly integrable and tight over E. If $f_n \to f$ a.e. on E, then f is integrable over E and

$$\lim_{n\to\infty} \int_E f_n = \int_E f$$

Proof: It is easy to verify that every f_n is integrable over E, since they are uniformly integrable and tight. So $|f_n|$ is integrable over E for every n. By Fatou's Lemma, $\int_E |f| \leq \liminf \int_E |f_n|$, and thus $|f|$ and f is integrable over E.

Now consider

$$(1) \quad \Big| \int_E f_n - \int_E f \Big| \leq \int_E |f_n - f| = \int_{E-E_0} |f_n - f| + \int_{E_0} |f_n - f|$$

, where E_0 is the corresponding subset of finite measure that meets the tightness challenge of $\frac{\epsilon}{4}$. So

$$(2) \quad \int_{E-E_0} |f_n| < \frac{\epsilon}{4}, \quad for\ every\ n$$

By Fatou's Lemma,

$$(3) \quad \int_{E-E_0} |f| \leq \liminf \int_{E-E_0} |f_n| < \frac{\epsilon}{4}$$

Finally, since E_0 has finite measure, by Vitali Convergence Theorem, there exists some N such that whenever $n \geq N$ we have

$$(4) \quad \int_{E_0} |f_n - f| < \frac{\epsilon}{2}$$

. Combining (1)-(4) we get the result as desired.

QED

Convergence in Measure

Let $\{f_n\}$ be a sequence of measurable functions defined on E. Let f_n, f be finite a.e. on E. We say that $f_n \to f$ **in measure** if for any η

$$m(\{x \in E \mid |f_n(x) - f(x)| > \eta\}) \to 0$$

It is easy to see that for any E, $Uniform\ Convergence \Rightarrow Convergence\ in\ Measure$.

If E has finite measure, then
$Uniform\ Convergence \Rightarrow Point-wise\ Convergence \Rightarrow Convergence\ in\ Measure$

Proposition 1.11.8: E has finite measure. If $f_n \to f$ a.e. on E, and f_n, f are finite a.e. on E, then $f_n \to f$ in measure.

Proof: Since $m(E) < \infty$, Egoroff's Theorem tells us that there is a subset E_0 such that $f_n \to f$ uniformly on E_0 and $m(E - E_0) < \epsilon$. So for sufficiently large n, the set $\{x \in E \mid |f_n(x) - f(x)| > \eta\} \cap E_0$ is empty, and thus $m(\{x \in E \mid |f_n(x) - f(x)| > \eta\}) \leq m(E - E_0) < \epsilon$.

QED

Proposition 1.11.9: If $f_n \to f$ in measure, then there is a sub-sequence $f_{n_k} \to f$ a.e. on E.

Proof: Since $f_n \to f$ in measure, we can construct an increasing sequence of n_k such that

$$E_k = \{x \in E \mid |f_{n_k}(x) - f(x)| > \frac{1}{k}\}$$

and $m(E_k) < \frac{1}{2^k}$. So $\sum_{k=1}^{\infty} m(E_k) < \infty$. By Borel-Cantelli Lemma, for almost every $x \in R$, x belongs to at most finitely many E_k. In other words, for every $x \in E$, there is some $K(x)$ such that when $k \geq K(x)$, $x \notin E_k$, i.e. $|f_{n_k}(x) - f(x)| \leq \frac{1}{K(x)}$. Therefore, $f_{n_k}(x) \to f(x)$ for almost every $x \in E$.

QED

Now we give three subsequence propositions stating the equivalence between the conditions of convergence and the exchange of $\lim$ and $\int$.

Proposition 1.11.10: Let E be of finite measure. Suppose $\{f_n\}$ is a sequence of non-negative functions such that $f_n \to f \equiv 0$ a.e. on E. Then, $\lim_{n\to\infty} \int_E f_n = 0$, if and only if, f_n is uniformly integrable over E .

Proof: If f_n is uniformly integrable over E, by Vitali Convergence Theorem, we have $\lim_{n\to\infty}\int_E f_n = 0$. Conversely, if $\lim_{n\to\infty}\int_E f_n = 0$, then for any $\epsilon > 0$ there exists some N such that when $n \geq N$ we have $\int_E f_n < \epsilon$. By Proposition 1.11.5, the finite sequence $f_1, \ldots, f_{N-1}$ is uniformly integrable. So there is some δ that meets the challenge of ϵ. As a result, the same δ meets the $\epsilon-$challenge for all f_n.

QED

Proposition 1.11.11: Suppose $\{f_n\}$ is a sequence of non-negative integrable functions such that $f_n \to f \equiv 0$ a.e. on E. Then, $\lim_{n\to\infty}\int_E f_n = 0$, if and only if, f_n is uniformly integrable and tight over E.

Proof: The right-to-left direction is true by applying General Vitali Convergence Theorem. For left-to-right, suppose $\lim_{n\to\infty}\int_E f_n = 0$. Note that it suffices to prove the tightness, because uniform integrability follows from the tightness. To see this, suppose f_n is tight over E. Then there is some E_0 of finite measure such that $\int_{E-E_0}|f_n| < \frac{\epsilon}{2}$. By Proposition 1.11.10, f_n is uniformly integrable over E_0. So there is some δ meeting the $\frac{\epsilon}{2}-$challenge in E_0. Thefore, for any $A \subseteq E$ such that $m(A) < \delta$, we have $A = A_1 \cup A_2$ where A_1, A_2 are disjoint and $A_1 \subseteq E_0$ and $A_2 \subseteq E - E_0$. So $\int_A |f_n| = \int_{A_1}|f_n| + \int_{A_2}|f_n| \leq \frac{\epsilon}{2} + \int_{E-E_0}|f_n| < \epsilon$.

To prove the tightness, since $\int_E f_n \to 0$, there exists some N such that when $n \geq N$ we have $\int_E f_n < \epsilon$. Note f_n is integrable over E for every n. By Lemma 1.11.6, there is some E_k meeting the tightness $\epsilon-$challenge for $f_k, k = 1, \ldots, N-1$. Take $E_0 = \bigcap_{k=1}^{N-1} E_k$. Then clearly E_0 meets the tightness $\epsilon-$challenge for $f_1, \ldots, f_{N-1}$. For $n \geq N$, we have $\int_{E-E_0}|f_n| \leq \int_E |f_n| < \epsilon$. Therefore, E_0 also meets the tightness $\epsilon-$challenge for f_n when $n \geq N$.

QED

Proposition 1.11.12: Suppose $\{f_n\}$ is a sequence of non-negative functions. Then, $\lim_{n\to\infty}\int_E f_n = 0$, if and only if, $f_n \to f \equiv 0$ in measure and f_n is uniformly integrable and tight over E.

Proof: For left-to-right, suppose $\lim_{n\to\infty}\int_E f_n = 0$. By General Vitali Convergence Theorem, f_n is uniformly integrable and tight over E (note in the proof of General Vitali Convergence Theorem, we don't require point-wise convergence of f_n). To show the convergence in measure, consider any η and by Chebyshev's inequality we have

$$m(\{f_n(x) \geq \eta\}) \leq \frac{1}{\eta}\int_E f_n$$

. So $0 = \lim_{n\to\infty} m(\{f_n(x) \geq \eta\}) = \lim_{n\to\infty}\frac{1}{\eta}\int_E f_n$.

Conversely, suppose $f_n \to f$ in measure and f_n is uniformly integrable and tight over E. Assume $\int_E f_n \to 0$ is not true. Then there is some ϵ_0 and a subsequence f_{n_k} such that

$$\int_E f_{n_k} \geq \epsilon_0$$

. Now by Proposition 1.11.9, since $f_n \to f$ in measure, there is a subsequence $f_{n_j} \to f$ a.e. on E. Clearly, f_{n_j} is also uniformly integrable and tight over E. By General Vitali Convergence Theorem, $\lim_{j\to\infty} \int_E f_{n_j} = 0$. The final step is to see that we may choose such a subsequence f_{n_j} such that f_{n_k} is a subsequence of f_{n_j}. And this leads to a contradiction.

QED

Characterizations of Riemann and Lebesgue Integrability

Lemma 1.11.13: Let $\{\phi_n\}$ and $\{\psi_n\}$ be sequences of functions that are integrable over E. In addition, ϕ_n are increasing and ψ_n are decreasing. Let f be a function that is defined on E, and

$$\phi_n \leq f \leq \psi_n, \ for\ all\ n\ on\ E$$

If

$$\lim_{n\to\infty} \int_E [\psi_n - \phi_n] = 0$$

, then $\phi_n \to f$ a.e. on E, $\psi_n \to f$ a.e. on E, f is integrable over E, and

$$\lim_{n\to\infty} \int_E \phi_n = \int_E f = \lim_{n\to\infty} \int_E \psi_n$$

Proof: Define $\phi(x) = \lim_{n\to\infty} \phi_n(x)$ for $x \in E$. Similarly, define $\psi(x) = \lim_{n\to\infty} \psi_n(x)$. Note that $\phi(x)$ and $\psi(x)$ are properly defined, since the monotone sequences $\phi_n(x)$ and $\psi_n(x)$ have limits (either a real value or ∞, $-\infty$). So,

$$\phi_n \leq \phi \leq f \leq \psi \leq \psi_n, \ for\ all\ n$$

. As a result, we have $0 \leq \psi - \phi \leq \psi_n - \phi_n$, and

$$(1) \quad \lim_{n\to\infty} \int_E (\psi - \phi) = 0$$

. (1) implies that $\psi - \phi = 0$ a.e. on E. Therefore, $\phi = f = \psi, a.e.\ on\ E$ and $\phi_n \to f$ a.e. on E and $\psi_n \to f$ a.e. on E. The remaining part is easy to prove.

QED

Proposition 1.11.14 (Sufficient and Necessary Conditions for Lebesgue Integral): Let f be a bounded function on a measurable set E. Then f is Lebesgue integrable, if and only if, f is a measurable function.

Proposition 1.11.15 (Sufficient and Necessary Conditions for Riemann Integral): Let f be a function on the closed interval $[a, b]$. Then f is Riemann Integrable, if and only if, f fails to be continuous at most on a set E with Lebesgue measure 0.